Annexure 03

Teaching & Examination Scheme and Syllabus

B. Tech Electrical Engineering

5th and 6th Semester

Batch – EE2019

w.e.f. ACY 2021-2022

Approved by BoS: March 2021

PANDIT DEENDAYAL ENERGY UNIVERSITY, GANDHINAGAR
SCHOOL OF TECHNOLOGY

COURSE STRUCTURE FOR B.TECH. IN ELECTRICAL ENGINEERING													
Semester VI			**B. Tech. in Electrical Engineering**										
Sr. No.	**Course/Lab Code**	**Course/Lab Name**	**Teaching Scheme**					**Examination Scheme**					
								Theory			**Practical**		**Total**
			L	**T**	**P**	**C**	**Hrs/Wk**	**MS**	**ES**	**IA**	**LW**	**LE/Viva**	**Marks**
11.	20EE311T	Power System Protection	3	0	0	3	3	25	50	25	--	--	100
12.	20EE311P	Power System Protection Laboratory	0	0	2	1	2	--	--	--	50	50	100
13.	20EE312T	Electric Drives	3	0	0	3	3	25	50	25	--	--	100
14.	20EE312P	Electric Drives Laboratory	0	0	2	1	2	--	--	--	50	50	100
15.	20EE3XXT	Professional Elective Course – II	3	0	0	3	3	25	50	25	--	--	100
16.	20EE3XXP	Professional Elective Course – II Laboratory	0	0	2	1	2	--	--	--	50	50	100
17.	20EE3XXT	Professional Elective Course – III	3	0	0	3	3	25	50	25	--	--	100
18.	20EE3XXP	Professional Elective Course – III Laboratory	0	0	2	1	2	--	--	--	50	50	100
19.	20EE3XXT	Open Elective - IV	3	0	0	3	3	25	50	25	--	--	100
20.	TP310	Industrial Training/ IEP (6 weeks)				2		--	--	--	--	100	100
		Total	**15**	**0**	**08**	**21**	**23**						**1000**

IA- Internal Assessment, MS-Mid Semester; ES – End Semester Exam

The students have to select two theory courses from the following basket for Professional Elective Course – II & III (Theory). Professional Elective Course – II & III Laboratory will be offered based on the selected Professional Elective Course – II & III (Theory).

Course Code	List of Professional Elective Course – II and III (Theory)	Course Code	List of Professional Elective Course – II and III Laboratory
20EE313T	Renewable Energy Engineering	20EE313P	Renewable Energy Engineering Laboratory
20EE314T	Advanced Power Electronic Converters	20EE314P	Advanced Power Electronic Converters Laboratory
20EE315T	Advanced Power System Analysis	20EE315P	Advanced Power System Analysis Laboratory
20EE316T	Embedded Systems	20EE316P	Embedded SystemsLaboratory

Department will offer Open Elective based on availability of expertise/faculty. This open elective will be offered to the students enrolled in B.Tech. programs of School of Technology other than Electrical Engineering.

Student of Electrical Engineering will have a choice for one open elective course from the basket of open electives announced by the other departments of School of Technology/other schools of university.

Course Code	Open Elective - IV
20EE317T	Electric Vehicles
20EE318T	Net Zero Energy Buildings

20EE311T					POWER SYSTEM PROTECTION					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To introduce the students with basic concepts of Relays, Protection schemes, Switch gear and Modern trends in protection for protecting the power system equipments
2. To appreciate and understand scientific concepts underlying engineering and technological applications
3. To educate the basic concepts and new developments in power system protection & Switchgear
4. To emphasize the significance of protection for electrical equipments

UNIT I BASICS OF POWER SYSTEM PROTECTION **10 Hrs.**

Introduction and Philosophy of A Protective Relaying System: Types of Faults, Abnormalities, Functions of Protective Relay Schemes, major Components of Power system, Basic Tripping Circuit, Testing and Maintenance of Relay, Zones of Protection, Requirements of Protective Systems, Relay Operating Criteria, Main and Backup Protection, Historical Review of Protective Relay Technology. **Protective Current And Potential Transformer:** CT Equivalent Circuit, Vector diagram, Construction, magnetization Curve, Core, Errors, accuracy, Specifications, Factors affecting selection PT: Equivalent circuit, Construction, CVT, Specifications. **Different Types of Relays: Electromagnetic Relays**: Classification, Thermal O/L Relays, Types Over Current Relays, Differential Relay, Directional Relay, Impedance Relays. **Static Relays:** Advantages and Limitations, basic Elements, Static Relays Architecture. **Microprocessor Based Digital Protection:** Advantages of Numerical Relays, Numerical Relay Hardware, Digital Signal Processing, Estimation of Phasors, Full Cycle Fourier Algorithm, Half Cycle Fourier Algorithm, Practical Consideration for Selection of Algorithm, DFT- FFT.

UNIT II EQUIPEMENT PROTECTION **14 Hrs.**

Generator Protection: Differential Protection , Inter-turn fault Protection, stator E/F, Rotor E/F, NPS, Field Failure, Over Load, Over Voltage, Reverse Power, Pole-Slipping, Back-up Impedance, Under Frequency , Miscellaneous Protection. **Transformer Protection:** Faults in Transformers, Gas operated relays, Over Current Protection, REF Protection, Differential Protection, Protection against over fluxing, Protection of Grounding transformers, Protection Against Overheating, Protection for small transformers. **Induction Motor Protection:** Starting of IM, Faults in IM, Abnormalities of IM, Protection of small IM, Protection of Large IM. **Numerical Approach to Apparatus Protection (Overview):** Generator Protection, Transformer Protection, Induction Motor Protection

UNIT III PROTECTION OF TRANSMISSION LINES **10 Hrs.**

Protection of Lines by Over Current Relays and Distance Relays, Carrier Current Protection for lines. **Bus Zone Protection:** Protection Requirements, Non unit protection, Unit protection schemes, Breaker Back-up Protection

UNIT IV THEORIES OF CIRCUIT INTERRUPTION & CIRCUIT BREAKERS **06 Hrs.**

Theory Of Circuit Interruption: Introduction, Physics of arc phenomena, Maintenance of the arc, Essential properties of arc, Arc interruption theories.**Important Terms, Phenomena's &Ratings related to circuit breakers:** Introduction, Arc-voltage, Re-striking Voltage, Recovery voltage, RRRV, Current chopping, Capacitive current breaking, Resistance Switching, Circuit Breaker Ratings. **Theory and Practice Of Conventional Circuit Breakers:** Classification of Circuit breakers, Types-Construction- Operating Principle- Relative merit-demerits of conventional circuit breakers (OIL Circuit Breaker, AIR Circuit Breaker)**Recent Developments In Circuit Breakers:**, Sulphur hexafluoride (SF6) circuit breaker, Vacuum circuit breaker, HVDC circuit breaker

Max. 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

- CO1 Understand physics of arc phenomena, basic theory of circuit interruptions, the construction and principle of operation for various types of circuit breakers
- CO2 Understand basic philosophy of protective relaying practices
- CO3 Compare electromagnetic, static and Microprocessor based relays
- CO4 Analyze protective relaying schemes for power system components
- CO5 Choose suitable protective relaying schemes for power system components
- CO6 Estimate relay settings for protective relaying schemes for power system components

TEXT/REFERENCE BOOKS

6. Oza, Nair, Mehta, Makwana, "Power System Protection and switchgear", TMH.
7. Stanley H. Horowitz, Arun G. Phadke, "POWER SYSTEM RELAYING"-Third Edition,2008 Research Studies Press Limited
8. J. Lewis Blackburn, Thomas J. Domin, "Protective Relaying Principles and Applications" –Third Edition, CRC Press, T&F, 2006.
9. C. Russell Masson, "Art And Science Of Protective Relaying",
10. Y. G. Parithankar and S. R. Bhide, "Fundamentals Of Power System Protection" 2nd edition, PHI
11. Bhavesh Bhalja, Nilesh Chothani, "Protection and switchgear", Oxford Publication 2011
12. B. Ram, "Power System Protection" TMH Publication

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100	**Exam Duration: 3 Hrs**
Part A/Question: 4 Questions, one from each unit, each carrying 15 marks	60 Marks
Part B/Question: 4 Questions, 1 from unit I and 2 from unit II, each carrying 10 marks	40 Marks

20EE311P					Power System Protection Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
-	-	2	1	2	-	-	-	50	50	100

Course Objectives:

1. To introduce the students with basic concepts of construction, operating Principle and working of various types of relays
2. To appreciate and understand scientific concepts underlying engineering and technological applications in the area of power system protections
3. To familiarise the students for relay setting procedures for various protection schemes
4. To emphasize & inculcate the significance of protection for electrical equipments through various performance based experiment through state of the art laboratory facilities

List of Experiments:

1. Introduction & familiarization with the laboratory
2. Study & Performance of MCB,ELCB,FUSE & plotting their performance characteristic
3. Study of the construction & Operation of Electromechanical Relay
4. Testing, Calibration of Electromechanical Over Current Relays (Normal Inverse, Very Inverse & Extreme Inverse Characteristics)
5. Principles of Radial feeder Protection – Calculations, Relay Settings
6. Principles of Radial feeder Protection –Verifications through Hardware Simulations
7. Study & familiarization of Numerical Relay
8. Principles of Over Voltage & Under voltage Protection – Calculations, Relay Settings
9. Principles of Over Voltage & Under voltage Protection – Verifications through Hardware Simulations
10. Principles of Parallel feeder Protection - Calculations, Relay Settings
11. Principles of Parallel feeder Protection– Verification through Hardware Simulations
12. Principles of Transformer Protection - Calculations, Relay Settings
13. Principles of Transformer Protection -Verification through Hardware Simulations
14. Principles of Transmission Line Protection (Distance Protection) – Calculations , Relay Settings & Verification through hardware simulations
15. Principles of Transmission Line Protection (Carrier Current Protection) – Calculations , Relay Settings & Verification through hardware simulations
16. Principles of Induction Motor Protection- Calculations , Relay Settings & Verification through hardware simulations
17. Principles of Generator Protection- Calculations & Relay Settings
18. Principles of Generator Protection -Verification through hardware simulation

COURSE OUTCOMES

On completion of the course, student will be able to;

CO1 – Understand the control & power circuit diagrams of Industrial grade panels
CO2 – Compare the performance characteristics of FUSE, MCB, and ELCB
CO3 – Understand construction, working, operation of Electromechanical / Electromagnetic, Microprocessor based / Numerical Relays
CO4 – Evaluate and validate relay settings for the over current protection, earth fault protections & distance protection of transmission lines
CO5 – Evaluate and validate relay settings for various protection schemes used for power transformers
CO6 – Evaluate and validate relay settings for various protection schemes used for rotating machines (Induction Motor, Alternator)

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100

Continuous evaluation	50 marks
End semester examination and Viva-voce	50 marks

20EE312T					Electric Drives					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To impart knowledge about the significance, fundamental concepts and elements of electric drives
2. To gain knowledge about the fundamental concepts and speed control of electric motors
3. To understand the operational analysis, applications, features and control of dc and ac motor drives

UNIT I: FUNDAMENTALS OF ELECTRIC DRIVES **08 Hrs**

Introduction to Electric Drives, Significance of Electric Drives, Elements of Electric Drives, Fundamental Torque Equation, Four Quadrant Operation, Classification of Load Torque, Steady State Stability, Basic Design of Electric Drives.

UNIT II: DC MOTOR DRIVES **04 Hrs**

Introduction to dc Motors, Starting and Braking of dc Motors, Conventional Techniques for Speed Control of dc Motors (Armature Voltage Control and Field Control), Solid State Control of dc Motors, Controlled Rectifier fed dc Motor Drives, Chopper fed dc Motor drives.

UNIT III: INDUCTION MOTOR DRIVES **14 Hrs**

Introduction to Induction Motors, Starting and Braking of Induction Motors, Impact of Unbalanced Stator Voltage on Motor Performance, Impact of Non-Sinusoidal Stator Voltage on Motor Performance, Conventional Techniques for Speed Control of Induction Motors (Stator Voltage Control, Pole Changing Method, Rotor Resistance Control, Frequency Control, Slip Power Recovery Scheme), ac Voltage Controller fed Induction Motor Drives, VSI fed Induction Motor Drives, Scalar Control, Static Rotor Resistance Control, Static Kramer Drive, Static Scherbius Drive.

UNIT IV: PERMANENT MAGNET BRUSHLESS DC & STEPPER MOTOR DRIVES **14 Hrs**

PMBLDC Motor Drives: Permanent Magnet Materials, Applications, Construction, Operating Principle & Characteristics, Electronic Commutation, Hall Effect Sensors for Position Measurement, Mathematical Modelling, Speed Control of PMBLdc Motors. **Stepper Motor Drives:** Application, Construction, Classification, Operating Principle and Characteristics, Power Converters for Stepper Motors, Stepper Motor Drives. Introduction to Linear Induction Motor Drives, Synchronous Reluctance Motor Drives and Switched Reluctance Motor Drives.

TOTAL HOURS 40 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

- CO1 – Understand the significance, structure and components of electric drives
- CO2 – Apply the fundamental concepts of dc motors for modelling and control of dc drives
- CO3 – Design the power and control structure for speed control of dc motor drives and analyze its operation under different operating scenarios
- CO4 – Apply the fundamental concepts of induction motors for analysing the classical speed control techniques
- CO5 – Develop the solid state control of induction motor and analyze its operation under various operating scenarios
- CO6 – Develop permanent magnet brushless dc and stepper motor drives after understanding the fundamental concepts related to the electric motors

TEXT/REFERENCE BOOKS

1. G. K. Dubey, "Fundamentals of Electrical Drives," 2nd Edition, Narosa Publication.
2. R. Krishnan, "Electric Motor Drives: Modelling Analysis and Control," Prentice Hall Inc.
3. B. K. Bose, "Modern Power Electronics and ac Drives," Prentice Hall Inc.
4. Vedam Subramanyam, "Electric Drives – Concepts and Applications," Tata McGraw Hill.
5. V.V. Athani, "Stepper Motors: Fundamentals, applications and Design", Tata McGraw Hill.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100	**Exam Duration: 3 Hrs**
Part A/Question: 4 Questions, one from each unit, each carrying 15 marks	60 Marks
Part B/Question: 4 Questions, 2 from unit III and 2 from unit IV, each carrying 10 marks	40 Marks

20EE312P					Electric Drives Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To impart the practical knowledge about the components and structure of electric drives
2. To learn operation of control and power architecture of dc drives and induction motor drives
3. To understand the operation of permanent magnet brushless dc and stepper motor drives and unique aspects related to the control architecture

LIST OF SIMULATIONS AND EXPERIMENTS

1. To simulate the mathematical model of dc motors.
2. To study the operation of controlled rectifier fed separately excited dc motor drive operating in open loop control and closed loop control mode through experimental studies
3. To study the operation of chopper fed separately excited dc motor drive operating in open loop control and closed loop control mode through experimental studies
4. To simulate controlled rectifier and chopper fed dc motor drives.
5. To study the operation of voltage source inverter fed 3-phase induction motor drive with sinusoidal pulse width modulation through experimental studies
6. To study the operation of voltage source inverter fed 3-phase induction motor drive with space vector modulation through experimental studies
7. To study the operation of V/f control of 3-phase induction motor through simulation studies
8. To simulate the VVVF control of 3-phase induction motor.
9. To simulate the static rotor resistance control of 3-phase induction motor.
10. To study the closed loop control of permanent magnet brushless dc motor through experimental studies.
11. To design and simulate the 3:6 decoder logic for the generation of gating signals for six inverter switches from three hall position sensor outputs
12. To simulate the permanent magnet brushless dc motor drives.
13. To study the operation of stepper motor drive through experimental studies.
14. To understand the operation of industrial drives.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – Understand the fundamental concepts, components and different types of electric drives
CO2 – Analyze power and control structures of electric drives
CO3 – Analyze the operation ac and dc drives through experimental studies
CO4 – Comprehend the unique features of control and power structure of permanent magnet brushless dc motor drives and stepper motor drives
CO5 – Operate industrial drives and evaluate their features and performance
CO6 – Validate the performance of electric drives through simulation studies

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100

Continuous evaluation	50 marks
End semester examination and Viva-voce	50 marks

20EE313T					Renewable Energy Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total/Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the concept of generation of renewable energy sources and distributed generation of power sources.
2. To acquire knowledge of basic types of available energy sources and connection of sources in to the power grid.
3. To acquire knowledge of Solar Photovoltaics and Wind energy systems and their control aspects.
4. To know the basics about other renewable energy technologies.

UNIT I INTRODUCTION TO THE ENERGY ENGINEERING **03 Hrs.**

Fundamentals and Classification of Energy Sources, Significance of Renewable Energy, Global and Indian Statistics, Energy Efficiency and Conservation, Indian Electricity Grid Code, Electricity Act2003, MNRE and National Smart Grid Mission.

UNIT II SOLAR ENERGYCONVERSION SYSTEM **15 Hrs.**

Solar Energy: Basic Definitions and Angles Related with Solar Energy, History, Harnessing Solar Energy, Solar Water Heating, Passive and Active Solar Cooling, Solar Thermal Electricity, Solar Photovoltaics, Solar Collectors. **Solar Photovoltaic (SPV) System:** Operating Principle and Construction of PV Cell, Technologies-Amorphous, Monocrystalline, Polycrystalline, Equivalent Circuit of Cell, Cells to Modules to Arrays, Standard Test Conditions (STC) of Photovoltaics, I-V &P-V Characteristics, Impacts of Temperature and Insolation, Shading Effect, Bypass and Blocking Diode, SPV Balance of System(BOS), Sizing and Designing A SPV System, Power Electronic Converters for SPV Systems, Maximum Power Point Tracking (MPPT) Algorithms, Battery Energy Storage System(BESS), Lead-Acid, Nickel-Cadmium and Lithium, Grid Connected and Off-Grid Topologies, Socio-Techno-Economic Analysis.

UNIT III WIND ENERGY CONVERSION SYSTEM **15 Hrs.**

Wind Energy Physics: Components of Wind Turbine Generating System and its Function, Betz Limit, Tip Speed Ratio, Active and Passive Stall and Pitch Control, Yaw control, Performance Co-efficient, Wind Power, Power-Speed &Power-Torque Characteristics, Wind Speed Statistics- Wind Energy Estimation, Discrete Wind Histogram, Wind Power Probability Density Functions, Weibull and Rayleigh Statistics. **Wind Generators**: Classification of Wind Turbines, Review of Modern Technologies, Types of Wind Energy Systems in Context with Type of Generator Used (Type A, B, C, D), Performance Calculations, Working of Fixed Speed Induction Generator, Effect of Capacitance, FSIG with STATCOM, Working and Operation of Doubly Fed Induction Generator, Control from Rotor Side and Grid Side Convertor, Comparison of Gear and Direct Drive Generators, Operation of Fully Rated Convertor Types of Generator (DDPMSG-Direct Drive Permanent Magnet Synchronous Generator, Basics of Standalone and Grid Connected Systems.

UNIT IV OTHERRENEWABLE ENERGY SOURCES **07 Hrs.**

Small Hydro Power (SHP): Definition & Classification (micro, mini & small), Types of SHP Plants: Components & Characteristics; **Tidal Power**: Concept, Tidal Turbine, Types of Tidal Power Plant, **Biomass, Geothermal & Fuel Cell**: Definition & Overview; **Grid-Level Storage**: Pumped Hydro, Compressed Air, Ultra capacitor & Super Magnetic Energy Storage (SMES); **Decentralized Energy Systems**: Concept, Microgrids.

TOTAL HOURS 40 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 – Understand the energy scenario and significance of power generation from renewable energy sources
CO2 – Define the basic terminologies involved in operation of solar and wind energy conversion systems
CO3 – Classify various types of wind energy generating system
CO4 – Analyze different topologies of solar energy conversion system
CO5 – Evaluate the performance characteristics of solar and wind energy systems
CO6 – Understand and introduce alternative energy sources for power generation

TEXT/REFERENCE BOOKS

1. Masters G., "Renewable and Efficient Electric Power Systems", John Wiley & Sons, Inc., Publication.
2. Lara O, Jenkins N, Ekanayake J, "Wind Energy Generation Modelling and Control", John Wiley & Sons, Ltd., 2009.
3. Mathew S., Wind Energy-Fundamentals, Resource Analysis and Economics, Springer, 2006.
4. Patel M R. "Design, Analysis, and Operation Wind and Solar Power Systems", Taylor & Francis, Second Edition.
5. Sawhney G. S., "Non-conventional energy sources", PHI Learning Pvt. Ltd.
6. Ackermann T., "Wind Power in Power Systems", John Wiley & Sons Ltd., 2005.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs**

Part A/Question: 4 Questions, one from each unit, each carrying 13 marks 52 Marks
Part B/Question: 4 Questions, 2 from unit II, 2 from unit III and 1 from unit IV, each carrying 12 marks 48 Marks

20EE313P					Renewable Energy Engineering Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
-	-	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

1. To understand and verify the solar PV characteristics for solar modules
2. To evaluate and analyse the performance of solar PV under different operating conditions
3. To gain practical knowledge of the operation and control of wind turbine generating system
4. To simulate hybrid energy system including energy storage element.

List of Experiments:

1. To study electricity bill of house/office/building/commercial complex/industry for last one year and analyse consumption pattern.
2. To demonstrate the I-V and P-V characteristics of PV module with varying radiation and temperature level with artificial solar PV kit and under natural sunlight.
3. To demonstrate the I-V and P-V characteristics of series and parallel combination of PV modules with artificial solar PV kit and under natural sunlight.
4. To demonstrate the effect of variation in tilt angle on PV module power.
5. To demonstrate the effect of shading on the output power in PV panel for single module, series connection and parallel connection.
6. To demonstrate the working of bypass diode and blocking diode for Solar PV module.
7. To work out power flow calculations of standalone PV system of DC load with battery.
8. To evaluate cut-in speed of wind turbine experimentally using wind energy training system.
9. To evaluate the Tip Speed Ratio (TSR) at different wind speeds using wind energy training system.
10. To evaluate the coefficient of performance of wind turbine using wind energy training system.
11. To plot the turbine power versus wind speed curve using wind energy training system.
12. To carry out simulation of PV module connected with buck converter fed to the DC load.
13. To carry out simulation of PV module connected with boost converter fed to the DC load.
14. To implement the maximum power point tracking algorithm in solar power plant.
15. To implement the maximum power point tracking in wind turbine generating system.
16. To develop and simulate an integrated hybrid model having different renewable energy sources and energy storage.
17. To estimate the wind forecast and wind potential.

COURSE OUTCOMES

On completion of the course, student will be able to;

CO1 – Understand the practical aspects of solar and wind energy sources
CO2 – Analyze the performance characteristics of solar and wind energy systems
CO3 – Simulate the operating concepts of solar photovoltaics and wind energy sources
CO4 – Implement the maximum power point tracking for solar and wind energy systems
CO5 – Evaluate the performance of solar and wind energy systems using relevant software
CO6 – Analyze the performance of hybrid energy systems

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100

Continuous evaluation	50 marks
End semester examination and Viva-voce	50 marks

20EE314T					Advanced Power Electronic Converters					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To give exposure to recent development in power semiconductor switches and power converters
2. To gain knowledge about the significance and power structure of advanced power converters
3. To analyse the operation and comprehend the applications of advanced power converters

UNIT I: POWER SEMICONDUCTOR SWITCHES& POWER CONVERTERS 06 Hrs

Review of Semiconductor Switches (Power Diode, SCR, DIAC, TRIAC, Power BJT, MOSFET, IGBT, GTO) and Power Converters, Wide Band Gap Power Semiconductor Switches, Heat Sink and EMI filter Design, Gate Drive ICs

UNIT II: MULTILEVEL INVERTERS & MULTI-PULSE RECTIFIERS 10 Hrs

Multilevel Inverters: Significance, Merits and Demerits, Power Quality and dv/dt rating, Classification, Operational Analysis of Neutral Point Clamped, Flying Capacitor and Cascaded Multilevel Inverters, Pulse Width Modulation for Multilevel Inverters, Applications. **Multi-pulse Rectifiers:** Power Quality Issues with 6-pulse Rectifiers, Significance of Multi-pulse Rectifiers, Operational Analysis of 12-Pulse, 18 Pulse and 24-Pulse Rectifiers, Applications, PWM rectifiers.

UNIT III: MATRIX CONVERTERS & RESONANT INVERTERS 10 Hrs

Matrix Converters: Features, Classification, Working of Direct, Indirect and Sparse Matrix Converters, Modulation Techniques, Application in Wind Energy Conversion System and Electric Drives. **Resonant Inverters:** Hard Switching, Effects of Hard Switching, Soft Switching, Resonant Inverters.

UNIT IV: SWITCHED MODE AND UNINTERRUPTIBLE POWER SUPPLIES 14 Hrs

Switched Mode Power Supplies: Operation of Buck and Boost dc-dc Converter with Discontinuous Current Conduction, Flyback Converter, Forward Converter, Push-Pull Converter, Half Bridge Converter, Full Bridge Converter, Single Active Bridge and Dural Active Bridge Converter, High Frequency Inductors and Transformers, Zero Voltage Switching, Zero Current Switching. **Uninterruptible Power Supplies:** Significance, Classification, Power and Control Architecture

TOTAL HOURS 40 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

- CO1 – Understand the power semiconductor switches, wide band gap switches and power converters
- CO2 – Design and select different components, such as heat sinks, EMI filters, gate driver ICs, required for the operation of power converters
- CO3 – Examine the power structure and operation of multilevel inverters and multi-pulse rectifiers
- CO4 – Classify different types of matrix converters and inspect their working and modulation strategies
- CO5 – Examine the operation of isolated dc-dc converter and design it as per the specified parameters
- CO6 – Analyze the power structure and operation of uninterruptible power supplies and resonant inverters

TEXT/REFERENCE BOOKS

1. M. H. Rashid, “Power Electronics: Circuits, Devices and Applications,” Prentice Hall of India Ltd., 2004.
2. Ned Mohan, T. M. Undeland and W. P. Robbins, “Power Electronics: Converters, Applications and Design,” Wiley India Ltd., 2003.
3. B. K. Bose, “Modern Power Electronics and ac Drives,” Prentice Hall Inc., 2001.
4. L. Umanand, “Power Electronics Essential & Applications,” Wiley India Pvt. Ltd., 2009.
5. D. W. Hart, “Power Electronics,” Tata McGraw-Hill, 2010.
6. Bin Wu, “High Power Converters and AC Drives,” Wiley - IEEE Press., 2017
7. Krishna Kumar Gupta and Pallavee Bhatnagar, “Multilevel Inverters: Conventional and Emerging Topologies and Their Control,” Academic Press, 2017.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs**

Part A/Question: 4 Questions, one from each unit, each carrying 15 marks 60 Marks

Part B/Question: 4 Questions, 1 from unit II, 1 from unit III and 2 from unit IV, each carrying 10 marks 40 Marks

20EE314P					Advanced Power Electronic Converters Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To impart the practical knowledge about working of modern power semiconductor switches and advanced power converters
2. To understand the features, merits, demerits and applications of advanced power converters
3. To analyze the operation of advanced power converters through experimental and simulation studies

LIST OF SIMULATIONS AND EXPERIMENTS

1. To study the application of power converters in different applications.
2. To implement gate driver circuitry for IGBT/MOSFET with the help of commercially available gate driver IC
3. To analyze the operation of wide band gap power semiconductor switches through experimental studies
4. To analyze the impact of EMI filters on the operation of power converters
5. To analyze the power quality at the input of the 6-pulse rectifier and determine its impact on another load connected at the point of common coupling
6. To demonstrate the operation of multilevel inverter and analyze the power quality at the output
7. To simulate the sinusoidal pulse width modulation for 5-level, 7-level and 11-level inverter
8. To simulate neutral point clamped multilevel inverter
9. To simulate cascaded multilevel inverter
10. To demonstrate the operation of 12-pulse rectifier and analyze the power quality at the rectifier input
11. To simulate and analyze the operation of multi-pulse rectifier
12. To design and analyze the operation of buck converter with discontinuous current conduction
13. To demonstrate the operation of isolated dc-dc converters
14. To simulate matrix converter and analyze its operation
15. To demonstrate the operation of uninterruptible power supply

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – Understand the significance of modern power semiconductor switches and advanced power converters
CO2 – Experiment with modern power semiconductor switches to determine their characteristics and features
CO3 – Design the gate drive circuitry for power semiconductor switches using commercially available driver ICs
CO4 – Experimentally validate the performance of the advanced power electronic converters
CO5 – Validate the performance of the advanced power electronic converters through simulation studies
CO6 – Analyze the operation of advanced power converters and determine their advantages with respect to conventional power converters

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100

Continuous evaluation	50 marks
End semester examination and Viva-voce	50 marks

20EE315T					Advanced Power System Analysis					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To introduce optimization techniques in power system to achieve optimal and secure operation of power systems
2. To introduce restructured power system and discuss issues in operating power system in new framework of deregulations
3. To explore methods of congestion management in restructured power system to achieve perfect power market

UNIT I: OPTIMAL POWER FLOW (OPF) **10 Hrs**

Economic dispatch (ED) background, difference between ED and OPF, OPF applications, OPF solution techniques, Gradient and Newton method for OPF.

UNIT II: RESTRUCTURED POWER SYSTEMS **08 Hrs**

Reasons for restructuring / deregulation of power industry, Understanding the restructuring process, Introduction to issues involved in deregulation, Reasons and objectives of deregulation of various power systems across the world, Consumer behaviour, Supplier behaviour, Market equilibrium, Relationship between short-run and long-run average costs, Perfectly competitive market, Market models based on contractual arrangements, Market architecture

UNIT III: POWER SYSTEM SECUIRTY AND CONGESTION MANAGEMENT **14 Hrs**

Functions of power security, optimal-secure dispatch, factors affecting security, derivation of generation shift factors and line outage distribution factors, contingency analysis using sensitivity factors, PTDF, contingency selection, congestion management, available transfer capability, locational marginal prices, nodal pricing, inter-zonal and intra-zonal congestion management

UNIT IV: POWER SYSTEM STATE ESTIMATIOM **08 Hrs**

Need of state estimation, least square methods for state estimation, state estimation of AC power network, state estimation solution algorithm, application of state estimation, derivation of Hessian Matrix

TOTAL HOURS 40 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

- CO1 – Operate power systems in most economical way by adopting optimization techniques considering various equality-inequality constraints and generation limits
- CO2 – Perform optimal and secure operation of interconnected power systems using optimal power flow techniques
- CO3 – Understand restructured power system and classify different market models, their architecture and mechanism
- CO4 – Formulate techniques for power system security analysis and apply sensitivity factors for contingency analysis
- CO5 – Manage congestion in restructured power system using different methods of ATC, OPF and nodal pricing
- CO6 – Estimate the state of power systems and explore its application to enhance system security

TEXT/REFERENCE BOOKS

1. Allen J Wood and Bruce F Woolenberg, “Power Generation, Operation and Control”, John Wiley & Sons
2. Hadi Sadat, “Power System Analysis”, Tata McGraw Hill
3. A R Abhayankar and S AKhaparde, “Restructured Power Systems”, Narosa Publications, 2011
4. J.D. Glover, M.S. Sharma and T.J. Overbye, “Power System Analysis and Design”, 6th Edition, Cengage Learning
5. John J. Grainger and William D. Stevenson, “Power System Analysis”, McGraw-Hill, 1994
6. J S Dhillon and D P Kothari, “Power System Optimization”, PHI Learning, 2004
7. NPTEL Web Course on Restructured Power Systems by Prof. S A Khaparde
8. N V Ramana, “Power System Operation and Control”, Pearson Education India, 2010

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs**

Part A/Question: 4 Questions, one from each unit, each carrying 15 marks — 60 Marks

Part B/Question: 4 Questions, 2 from unit I, 1 from unit II and 2 from unit III, each carrying 10 marks — 40 Marks

20EE315P					Advanced Power System Analysis Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To introduce optimization techniques for power system and use of simulation software
2. To learn programming and use of simulation software for economic dispatch (ED), security analysis and congestion management
3. To explore the issues in power system security and find their solutions

LIST OF SIMULATIONS

1. To solve constrained parameter optimization with equality constraints using Newton's method
2. To solve ED problem of thermal generating units using lambda iteration method
3. To solve ED problem of thermal generating units considering generation limits and transmission line losses
4. To simulate Optimal Power Flow (OPF) for interconnected power systems in simulation software
5. To simulate and compare multi-area optimization problem as ED, OPF and Security Constrained Optimal Power Flow (SCOPF)
6. To simulate contingency analysis considering single/multiple outage of line or generators from interconnected power system
7. Formulation of LODF, PTDF and Generation Shift Factors for Contingency Analysis
8. To use locational marginal price (LMP) to identify congestion in power systems
9. To simulate available transfer capability for congestion management in power system
10. To carry out state estimation of power system

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – Operate simulation software to analyze power systems operations
CO2 – Write programs for power system optimization
CO3 – Simulate interconnected power system for ED-OPF and differentiate both solutions
CO4 – Solve contingency analysis and apply sensitivity factors to identify different state of power system operation
CO5 – Assess congestion management using different analysis techniques
CO6 – Solve state estimation problem in power system

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100

Continuous evaluation 50 marks
End semester examination and Viva-voce 50 marks

20EE316T					Embedded Systems					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To impart knowledge about the concept, significance and applications of embedded systems
2. To gain knowledge about the technology involved in embedded systems
3. To comprehend the operation of microcontrollers, its peripherals and interfaced devices programmed as per the need of the application

UNIT I: INTRODUCTION TO EMBEDDED SYSTEMS **10 Hrs**

Embedded Systems – Definition, Significance, Classification, Features, Examples and Applications, Basic Architecture of Microcontrollers (RISC and CISC architecture, Flash, SRAM, EEPROM and Solid State Memories), Inside the Embedded Systems.

UNIT II: MICROCONTROLLER ARCHITECTURE **10 Hrs**

Popular Platforms, Device Overview, Features, Functional Block Diagram, Pin Diagram, Specifications, CPU, Memory and Bus Architecture, Flash Memory, Clock, Oscillator and Phase Locked Loop, General Purpose Input Output Registers, General and Specific Register Sets, DMA Controller, Time Delays, Pipeline, Board Power Management, Programming the Microcontroller.

UNIT III: MICROCONTROLLER PERIPHERALS **10 Hrs**

Analog to Digital Converter, Digital to Analog Converter, Timers, Counters (Up Counter, Down Counter, Up-Down Counter), Watchdog Timer, Interrupts, Capture, Pulse Width Modulation Control (Synchronous and Asynchronous Modulation), Quadrature Encoder Pulse Circuitry, Basics of Serial and Parallel Communication, Serial Peripheral Interface (SPI), USART, Inter-integrated Communication, USB Communication, Enhanced Controller Area Network, Programming Microcontroller Peripherals.

UNIT IV: INTERFACING AND APPLICATIONS **10 Hrs**

Interfacing LEDs, Relays, Buzzers, Sensors (Current Sensors, Voltage Sensors, Speed and Positions Sensors, Light Sensors, Temperature Sensors, Proximity Sensors, etc.). Programming Microcontroller with peripherals and interfaced devices. Application in Control of Electric Drives, Electric Vehicles, etc.

TOTAL HOURS 40 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – Illustrate the features, components and applications of embedded systems
CO2 – Understand the architecture and functionality of microcontrollers
CO3 – Interpret the hardware and software design requirements of the embedded system
CO4 – Apply different peripherals and interface devices as per the need of the applications
CO5 – Gain the ability to code an embedded system to facilitate the desired operation
CO6 – Design a solution for specific application using embedded system

TEXT/REFERENCE BOOKS

1. K. V. Shibu, **"Introduction to Embedded Systems,"** McGraw Hill Publication, 2013.
2. Frank Vahid and Tony Givargis, **"Embedded System Design,"** Wiley Publication, 2002.
3. Steve Heath, **"Embedded Systems Design,"** 2nd Edition, Newnes Publication.
4. Wayne Wolf, **"Computers as Components: Principles of Embedded Computing System Design,"** 2nd Edition, Morgan Kaufmann Publishers, 2008.
5. Application Notes and Reference Manuals

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs**

Part A/Question: 4 Questions, one from each unit, each carrying 15 marks 60 Marks
Part B/Question: 4 Questions, one from each unit, each carrying 10 marks 40 Marks

20EE316P					Embedded System Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To impart the practical knowledge about the components and structure of embedded systems
2. To understand about the microcontroller architecture, peripherals and interfacing
3. To learnabout the operation and coding of embedded systems

LIST OF SIMULATIONS AND EXPERIMENTS

1. Blink an LED connected to GPIO pin configured as an output pin of microcontroller
2. Accept input through a push button switch connected to a GPIO pin configured as an inputof microcontroller
3. Interface an LCD with a microcontroller todisplay the Integers
4. Control an output voltage at DAC using potentiometer connected to the ADC of the microcontroller
5. Display floatingnumbers on an LCD with microcontroller
6. Perform A to D Conversion and measure the data using DAC
7. Interface a temperature sensor with the ADC and display the temperature
8. Use Timers Counters and the Prescaler on the microcontroller
9. Demonstrate Up, Down and Up-Down counter operation using a microcontroller
10. Demonstrate a PWM (Pulse Width Modulation) using a microcontroller
11. Control the PWM Output with the ADC and a potentiometer

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – Understand components and features of an embedded system
CO2 – Understand the architecture, peripherals and interfacing of microcontroller
CO3 – Analyze the operation an embedded system through experimental studies
CO4 – Interpret the hardware and software design requirements of the embedded system
CO5 – Code a microcontroller to implement the desired task
CO6 – Design an embedded system for a specific application

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100

Continuous evaluation	50 marks
End semester examination and Viva-voce	50 marks

20EE317T					Electric Vehicles					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To give exposure to electric vehicle technology and government policies for electric vehicles
2. To gain knowledge about the operation of electric powertrain and charging infrastructure
3. To analyse the application of artificial intelligence, machine learning and embedded systems in electric vehicles

UNIT I: ELECTRIC VEHICLES **09 Hrs**

Concept, Significance, Classification, Components, Vehicle Dynamics, Merits and Demerits, Challenges and Constraints, Power Train, Charging Requirements, V2G and G2V technology, Technical Specifications of Commercial Electric Vehicles, Government Policies on Electric Vehicles

UNIT II: ELECTRIC POWERTRAIN **11 Hrs**

Concept, Classification, Power Train Architecture, Components, Propulsion Motors, Permanent Brushless dc Motors, Sensors, Power Converters, Controller, Power and Control Architecture, Battery Management System, Regenerative Braking, Drive Cycle and Testing, 2-Wheeler and 4-Wheeler Electric Vehicle, Hybrid Electric Vehicle Drive Trains

UNIT III: CHARGING INFRASTRUCTURES **11 Hrs**

Energy Storage Elements, Battery, Fuel Cell, Ultracapacitors, Mathematical Modelling of Energy Storage, Hybrid Energy Storage, Battery Charging, Battery Chargers, Power Converters for Battery Charging, Constant Voltage, Constant Current and Boost Charging, Impact of Battery Charging on Grid, Solar Photovoltaic Based Battery Charging, Off-board, On-board and Wireless Battery Charging, Vehicle to Grid and Grid to Vehicle Interface Elements

UNIT IV: EMERGING TRENDS IN ELECTRIC VEHICLES **09 Hrs**

Wide Band Gap Semiconductor Devices, Application of Artificial Neural Networks, Fuzzy Logic, Machine Learning and IoT in Electric Vehicles, Communication Systems, Integration of Electric Vehicles in Smart Grids.

TOTAL HOURS 40 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – Understand the electric vehicle technology, its significance and government initiatives for electric vehicles
CO2 – Classify different types of electric vehicles and list out the components being used in electric powertrain
CO3 – Examine the power and control architecture of electric powertrain
CO4 – Classify the different energy storage elements employed in electric vehicles
CO5 – Analyze the operation of charging infrastructure and battery management systems
CO6 – Explore the emerging trends in electric vehicle technology and understand the application of artificial intelligence and IoT in electric vehicles

TEXT/REFERENCE BOOKS

1. Mehrdad Ehsani, Yimi Gao, Sebastian E. Gay, and Ali Emadi, "Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design," CRC Press, 2004.
2. Iqbal Hussein, "Electric and Hybrid Vehicles: Design Fundamentals," CRC Press, 2003.
3. B. K. Bose, “Modern Power Electronics and ac Drives,” Prentice Hall Inc., 2001.
4. Sheldon S. Williamson, “Energy Management Strategies for Electric and Plug-in Hybrid Electric Vehicles,” Springer, 2013.
5. C.C. Chan and K.T. Chau, “Modern Electric Vehicle Technology,” Oxford University Press, 2001.
6. Rodrigo Garcia-Valle and João A. Peças Lopes (Eds.), “Electric Vehicle Integration into Modern Power Networks,” Springer, 2012.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs**

Part A/Question: 4 Questions, one from each unit, each carrying 15 marks 60 Marks
Part B/Question: 4 Questions, one from each unit, each carrying 10 marks 40 Marks

20EE318T					Net Zero Energy Buildings					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	-	-	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. Exposure to the concept of Net zero energy (nZEB) and Green buildings.
2. Understand the end-use consumption pattern in buildings
3. Evaluate comfort calculation for occupants in buildings
4. Application of Renewable energy systems and storage to develop a Nzeb

UNIT I: INTRODUCTION **07Hrs.**

Building energy consumption: Global energy and environment scenario, Building energy consumption, What is nZEB?, Why *Zero*?, Energy Units &Conversions, Definitions, Site and source energy, Energy Basis vsGHG Basis, Building Energy Systems, Energy performance indices. Concept of Green Buildings and Ratings.

UNIT II: ENERGY USAGE IN BUILDINGS **10Hrs.**

Energy use intensity, End uses, Energy transfer processes in building space, Building energy modelling and simulation: Building envelope model-Thermal network model, Resistance-Capacitance (RC) network, transfer function method, single zone model of building space, Heating/cooling load calculation HVAC and Lighting systems, Principles for Net Zero Building Enclosures, Passive and Active Systems

UNIT III: OCCUPANCY COMFORT ANALYSIS **05 Hrs.**

Interactions between forms of comfort and building energy use, Thermal comfort, Visual comfort and daylighting, Indoor air-quality, acoustic comfort, Occupants; comfort calculation

UNIT IV: RENEWABLE ENERGY SYSTEMS FOR BUILDINGS **11Hrs.**

Review of renewable energy technologies, BiPV,BiPV/T,BaPV, Solar thermal systems with storage, Solar air-conditioning, combined heat and power, Energy storage: Batteries-Lithium, Lead-acid, Thermal energy storage-Phase change materials, hydrogen. Geothermal.Smart buildings: Grid interactive buildings,Energy management-Electric vehicles, *Vehicle-Building-Grid* integration, smart homes, positive energy buildings

UNIT V BUILDING ENERGY OPTIMIZATION **07 hrs.**

Review of energy optimization techniques, Building energy usage calculation, Occupants' behaviour based calculation, Cost-optimality, nearly-nZeB, Case studies, Home energy management systems, Smart homes for rural areas, and future scope.

TOTAL HOURS 40 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 – Understand the concept of net zero energy building and modelling methods
CO2 – Understand energy transfer processes within and for a building space
CO3 – Apply optimization techniques to optimize the energy consumption in buildings
CO4 – Apply renewable energy technologies for improve energy performance of buildings
CO5 – Examine the energy systems responsible for energy usage in buildings
CO6 – Determine optimal measures to save energy in a building

TEXT/REFERENCE BOOKS

1. Athienitis, Andreas K., and William O'Brien, eds. "Modelling, design, and optimization of net-zero energy buildings". Berlin: Ernst &Sohn, 2015.
2. Attia, Shady. "Net Zero Energy Buildings (NZEB): Concepts, frameworks and roadmap for project analysis and implementation". Butterworth-Heinemann, 2018.
3. Jadhav, Nilesh Y. "Green and smart buildings: advanced technology options". Springer, 2016.
4. Parkin, Robert E. "Building-integrated solar energy systems". CRC Press, 2017.

WEBLINKS

1. World Green buildings Council.www.worldgbc.org, Indian Green Building Council. https://igbc.in/igbc/
2. TERI.https://www.teriin.org/buildings.
3. Bureau of Energy Efficiency (BEE) India. https://beeindia.gov.in/

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs**

Part A/Question: 5 Questions, one from each unit, each carrying 12 marks 60 Marks
Part B/Question: 4 Questions , one each from unit I, II, IV, and V, each carrying 10 marks 40 Marks

TP310					Industrial Training/ International Exposure Program					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
-	-	-	2	6 weeks	--	--	--	--	100	100

COURSE OBJECTIVES

1. To provide students an exposure to the working environment of the industry
2. To understand the products being developed and/or services being offered by the industry
3. To cultivate the necessary technical report writing and oral presentation skills

Students are required to carry out 6 weeks of industrial internship in industries to get an exposure to the working of an industry. The students will have the opportunity to study various industrial processes, departments involved and how a product is manufactured. Moreover, the students will get an opportunity to see the theoretical knowledge acquired being put into practice in industry. They will also come to know about the technical and inter-personal skills for working in an industrial environment.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 – Understand the working of industry
CO2 – Comprehend the technical skills required while working in the industry
CO3 – Apply the technical knowledge in industrial scenario
CO4 – Examine the products being developed or services being offered by the industry
CO5 – Examine the process being employed for developing the products or services
CO6 – Acquire the necessary skills for technical report writing

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100

Presentation and Viva-voce 100 marks